

Convexity, Hessians and Newton's method

Chords, second derivatives and the Hessian test for convexity; why least squares and cross-entropy pass it.

THE QUESTION

When is the minimum we find the best one, and can curvature get us there in a few steps?

THE INTUITION

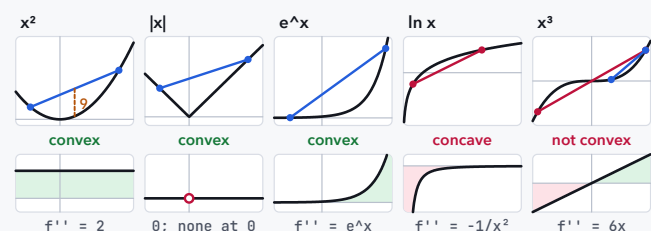
A convex function lies on or below its chords, so it has no false valleys: any local minimum is global. The second derivative, or the Hessian, tests it. Newton's method uses the same curvature to jump to the bottom of a local quadratic.

KEY TAKEAWAY

Convex $\Leftrightarrow f'' \geq 0$, or H PSD, everywhere. Least squares and cross-entropy are convex. Newton: $\theta \leftarrow \theta - H^{-1}g$; for logistic regression that is IRLS.

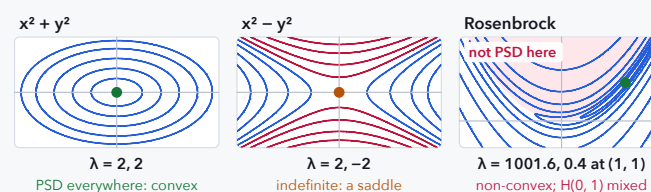
chords above $\rightarrow f'' \geq 0, H$ PSD $\rightarrow$ local min is global $\rightarrow$ local quadratic $\rightarrow \theta - H^{-1}g \rightarrow$ IRLS

1 Chords lie above the graph, and $f'' \geq 0$



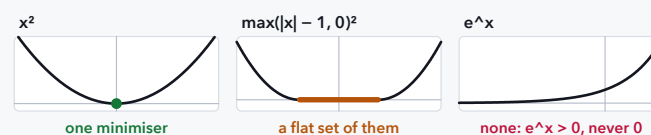
f is convex on a set if $f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2)$ for all x_1, x_2 in it and $t \in [0, 1]$: no chord dips below the graph. For x^2 , LHS - RHS = $(t^2 - t)(x_1 - x_2)^2 \leq 0$ as $t^2 \leq t$; from -2 to 4 the gap peaks at 9 . f is concave if $-f$ is convex, like $\ln x$. A twice-differentiable f is convex on an interval exactly when $f'' \geq 0$ there (bottom row): the slope never falls. $f'' > 0$ is enough but not needed: x^4 is strictly convex with $f''(0) = 0$. x^3 ($f'' = 6x$) is convex only for $x \geq 0$. $|x|$ has no f'' at 0 but passes the chord test.

2 Many parameters: the Hessian must be PSD



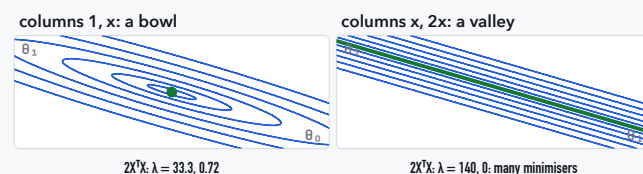
$H_{ij} = \partial^2 f / \partial \theta_i \partial \theta_j$, symmetric when these are continuous. f is convex exactly when H is positive semidefinite (PSD) at every point: $v^T H v \geq 0$ for every direction v (the curvature along v), or all eigenvalues ≥ 0 . For $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$, $\lambda = (\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det})/2$. Rosenbrock's $(1-x)^2 + 100(y-x^2)^2$ has $\lambda = 1001.6$ and 0.4 at its minimum $(1, 1)$: a valley 2500 times steeper across than along. At $(0, 1)$ they are -398 and 200 ; H is indefinite wherever $y > x^2 + 0.005$. At a single point with $\nabla f = 0$, all $\lambda > 0$ means a local minimum, all $\lambda < 0$ a maximum, and mixed signs a saddle ($x^2 - y^2$ at 0). The size of λ is the steepness along its eigenvector.

3 Any minimum is global, if one exists



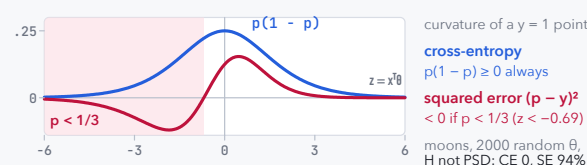
All $\lambda \geq 0$ everywhere: every local minimum is global, but there may be a flat set of them. All $\lambda > 0$: strictly convex, at most one minimiser. Neither promises a minimum: e^x has none, nor has cross-entropy when a line separates the classes.

4 Least squares is convex: $H = 2X^T X$



$\|y - X\theta\|^2 = y^T y - 2y^T X\theta + \theta^T X^T X\theta$ has gradient $-2X^T y + 2X^T X\theta$ (the maths for ML sheet) and $H = 2X^T X$ at every θ . As $v^T H v = 2\|Xv\|^2 \geq 0$, it is PSD for any X . If one column is a multiple of another, $Xv = 0$ for some $v \neq 0$: $\lambda = 0$ and a line of minimisers. kf with $k \geq 0$ and sums of convex functions are convex, so ridge ($+\lambda\theta^T\theta$ adds 2λ to each eigenvalue) and lasso are too.

5 Cross-entropy is convex: $H = X^T S X$

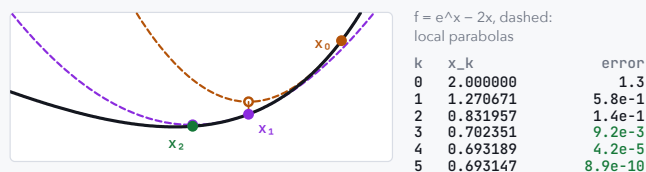


Differentiating the logistic regression sheet's gradient $X^T(p - y)$, $p = \sigma(X\theta)$, once more gives $H = \sum_i p_i(1-p_i)x_i x_i^T = X^T S X$ with $S = \text{diag}(p_i(1-p_i))$. So $v^T H v = \sum_i s_i (x_i^T v)^2 \geq 0$ at every θ , and > 0 if the columns of X are independent. Squared error through the sigmoid gives a $y = 1$ point the weight $2p(1-p)^2(3p-1)$, negative when $p < 1/3$, so it is not convex: on the moons its Hessian fails PSD at 94% of 2000 random θ , cross-entropy's at none.

Newton steps to the bottom of a local quadratic

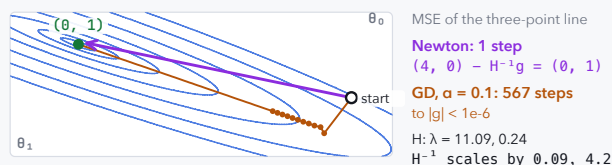
The Newton step, one step on a quadratic, weighted least squares, IRLS, and step counts against gradient descent.

6 Newton's step minimises the local quadratic



Near θ_k , $f \approx f(\theta_k) + g_k^T \Delta + \frac{1}{2} \Delta^T H_k \Delta$ with $\Delta = \theta - \theta_k$, $g_k = \nabla f(\theta_k)$, $H_k = \nabla^2 f(\theta_k)$. Its gradient $g_k + H_k \Delta$ is 0 at $\theta_{k+1} = \theta_k - H_k^{-1} g_k$; in 1-D, $x - f'(x)/f''(x)$. On $e^x - 2x$ (illustrative) the error roughly squares each step near the minimum: quadratic convergence. Gradient descent cuts it by a constant factor.

7 On a quadratic one Newton step is exact



A quadratic is its own quadratic model. On the gradient descent sheet's line through (1, 1), (2, 2), (3, 3) the MSE has $H = \begin{pmatrix} 2 & 4 \\ 4 & 9.33 \end{pmatrix}$ and $g = (4, 6.67)$ at (4, 0), and one step lands on the minimum (0, 1); gradient descent with $\alpha = 0.1$ takes 567. H^{-1} divides the step along each eigenvector by its eigenvalue: a learning rate per direction, none to tune.

8 Weighted least squares in closed form



Give point i a weight $w_i \geq 0$ and minimise $(y - X\theta)^T W (y - X\theta)$, $W = \text{diag}(w_i)$. The gradient $-2X^T W (y - X\theta) = 0$ gives $\theta = (X^T W X)^{-1} X^T W y$, the minimum since $2X^T W X$ is PSD. $W = I$ is ordinary least squares. Weight 0.1 on the doubtful last point pulls the line back to the other four.

11 Pause and think

Non-convex f ? Newton may head for a saddle or a maximum.
 Is kf convex? Only for $k \geq 0$.

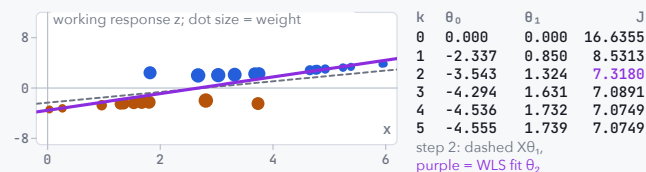
Separable classes? $S \rightarrow 0$; add $\lambda \theta^T \theta$.
 Is $|x|$ convex? Yes, by chords.

Learning rate? None: H^{-1} sets the step.
 Large d ? $O(d^3)$ a step: use GD.

convex: $f'' \geq 0$, H PSD · minima are global · $2X^T X$, $X^T S X$ PSD · Newton $\theta - H^{-1}g$ is IRLS

Before: the logistic regression sheet; next: the precision-recall curves sheet. Source: lecture slides "Convex functions", "Convexity, Hessian, IRLS" and the Hessian, Newton and IRLS part of "Logistic regression", and the second-order logistic regression notebook (nipunbatra.github.io/ml-teaching). The moons use our seed; the 1-D examples and weighted points are ours. Corrected: the f'' test is ≥ 0 , a convex function need not have a minimum, and $X^T S X$ is positive definite only when the columns of X are independent.

9 IRLS: Newton as weighted least squares



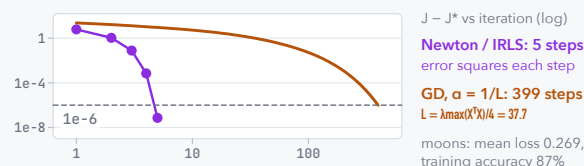
Put $g = X^T(p - y)$ and $H = X^T S X$ into Newton's step:

$$\theta_{k+1} = \theta_k + (X^T S_k X)^{-1} X^T (y - p_k) = (X^T S_k X)^{-1} X^T S_k z_k$$

with working response $z_k = X\theta_k + S_k^{-1}(y - p_k)$: weighted least squares with $W = S_k$, redone as both change ("iteratively reweighted"). Points with p_i near 0.5 weigh most. On 24 illustrative points it settles in 5 steps and matches Newton to 10^{-14} :

```
def irls(X, y, iters=8):
    th = np.zeros(X.shape[1])
    for _ in range(iters):
        p = 1 / (1 + np.exp(-X @ th))
        s = p * (1 - p) # weights
        z = X @ th + (y - p) / s # z_k
        XtS = X.T * s # X^T S
        A, b = XtS @ X, XtS @ z
        th = np.linalg.solve(A, b)
    return th
```

10 Few Newton steps, each $O(nd^2 + d^3)$



The second-order notebook's two moons (100 points, noise 0.1, our seed). From $\theta = 0$ Newton is within 10^{-6} of the minimum in 5 steps (the notebook: 5, on its own draw); gradient descent with the safe step $1/L$, $L = \lambda_{\max}(X^T X)/4$, needs 399. Each Newton step costs $O(nd^2 + d^3)$ time and $O(d^2)$ memory: solve $H\delta = -g$ (Cholesky) rather than invert, and keep it for small to medium d .